

2-DOF Robotic Ankle Joint Actuator Sizing Report

Complete torque evaluation of three Robstride motors (RS02, RS03, RS04) across four mechanism configurations for a 2-degree-of-freedom humanoid ankle joint. Each configuration is analyzed at every joint angle to determine worst-case torque capacity for both continuous operation and peak impact loading.

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Prepared For: Ankle Actuator Selection & Design Review
Robot System: 2-DOF Parallel Linkage Ankle (2-RSS Configuration)
Motors Evaluated: Robstride RS02 (6 N·m rated / 17 N·m peak), RS03 (20 / 60), RS04 (40 / 120)
Configurations: Standard Parallel × 2 spread widths + Orthogonal × 2 spread widths
Date: July 10, 2026

1. How the Ankle Mechanism Works

1.1 What is a 2-DOF Ankle?

The ankle joint rotates around two axes: **Pitch** (tilting the foot forward/backward, like pressing a gas pedal) and **Roll** (tilting the foot sideways, like standing on the edge of your foot). These two combined let the robot balance on uneven ground.

1.2 What Drives It?

Two rotary servo motors sit inside the robot's lower leg. Each motor has a short arm (the **crank**) attached to its output shaft. A long rod (the **pushrod**) connects the end of each crank arm down to the foot platform through a universal joint (U-Joint). When a motor spins its crank, the pushrod pushes or pulls the foot, tilting it. The two motors working together control both pitch and roll.

1.3 Key Dimensions

LABEL	FULL NAME	WHAT IT MEANS
r ₁ , r ₂	Crank 1 / Crank 2 Radius	Length of the motor's arm. Bigger = more leverage but needs more space.
L ₁	Pushrod 1 Length	Length of the connecting rod from Motor 1's crank to the foot.
L ₂	Pushrod 2 Length	Length of the connecting rod from Motor 2's crank to the foot.
H	U-Joint to Motor 1 Height	Vertical distance from ankle pivot up to the top motor (Motor 1).
D	U-Joint to Motor 2 Height	Vertical distance from ankle pivot up to the bottom motor (Motor 2).
G	Foot-to-UJoint Height	Height of the ankle pivot point above the bottom of the foot plate.
E	Ankle Spread	Horizontal gap between the two pushrod attachment points on the foot. Wider = better roll torque.
L	Motor Depth Y-Offset	How far forward/backward the motors are offset from the ankle center.
h	Attachment Offset	Vertical offset of pushrod attachment on the foot above/below the U-Joint.

1.4 How Motor Torque Converts to Ankle Torque

Motor torque does not transfer 1:1 to the ankle. The linkage geometry amplifies or reduces it depending on the foot angle. We use the **Jacobian matrix** (a 2×2 matrix that changes at every foot position) to compute exactly how much ankle torque results from a given motor torque:

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τ_ankle = JT × τ_motor    |    J is the 2×2 Jacobian at the current (Pitch, Roll) position
For Pitch torque capacity: τ_pitch = (|J11| + |J21|) × τ_motor    (both motors contributing to pitch)
For Roll torque capacity: τ_roll = (|J12| + |J22|) × τ_motor    (both motors contributing to roll)
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The **worst-case minimum** across all joint angles is what matters. If at any angle the torque drops too low, the robot falls.

1.5 How the Evaluation Sweep Works

Pitch Sweep: Fix Roll = 0°. Move Pitch from -25° to +25° in 1° steps. At each step, compute ankle pitch torque using the Jacobian. Record the minimum value and the angle where it occurs.

Roll Sweep: Fix Pitch = 0°. Move Roll from -40° to +40° in 2° steps. At each step, compute ankle roll torque using the Jacobian. Record the minimum value and the angle where it occurs.

2. The Four Configurations

Standard Parallel Layout

Both motors face the same direction (rotation axes parallel). When the foot pitches, both motors cooperate (push together). When the foot rolls, motors oppose each other (one pushes, one pulls). Result: **strong pitch, weak roll** — especially with narrow ankle spread.

Orthogonal Layout (Top Motor Rotated 90°)

Motor 1 (top) is rotated 90° in yaw so its crank sweeps side-to-side. This separates the motors' action planes: Motor 1 primarily controls roll, Motor 2 primarily controls pitch. Result: **balanced pitch and roll** — decoupled axes.

Ankle Spread E = 2.3 cm (Current Design)

Narrow gap between pushrod attachment points. Gives the motors very little lateral leverage for roll. This is the existing robot design.

Ankle Spread E = 5.0 cm (Parallel Design)

Wider gap between attachment points. Directly increases roll torque by providing a wider base for the push-pull action.

The 4 Configurations:
Config 1: Standard + E=2.3 cm (current robot) | **Config 2:** Standard + E=5.0 cm (widened)
Config 3: Orthogonal + E=2.3 cm (decoupled, compact) | **Config 4:** Orthogonal + E=5.0 cm (decoupled, wide)

3. Motor Specifications

MOTOR	CONTINUOUS RATED TORQUE	PEAK IMPACT TORQUE	GEOMETRY USED
RS02	6.0 N·m	17.0 N·m	r=2.5, L ₁ =24.3, L ₂ =16.3, H=23.5, D=16.0, G=3.6, L=3.8 cm
RS03	20.0 N·m	60.0 N·m	r=2.5, L ₁ =22.2, L ₂ =11.2, H=22.2, D=11.2, G=3.6, L=3.8 cm
RS04	40.0 N·m	120.0 N·m	r=2.5, L ₁ =22.2, L ₂ =11.2, H=22.2, D=11.2, G=3.6, L=3.8 cm

RS02 uses the current base dimensions from the existing robot. RS03 and RS04 use updated dimensions from new physical measurements.

4. Overall Sizing Verdict — Master Performance Matrix

Guaranteed worst-case minimum ankle torque across the Pitch [−25°, +25°] and Roll [−40°, +40°] workspace limits, alongside the critical pose angles where the minimum values occur.

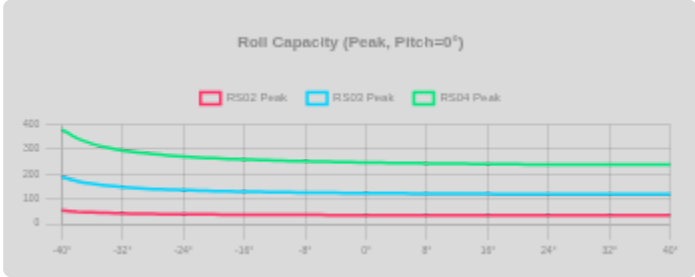
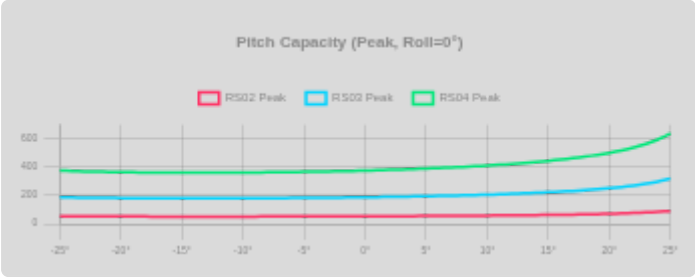
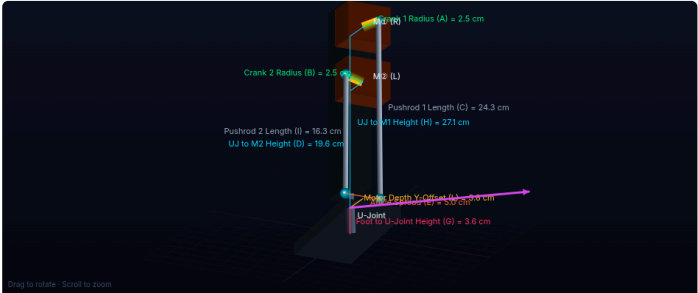
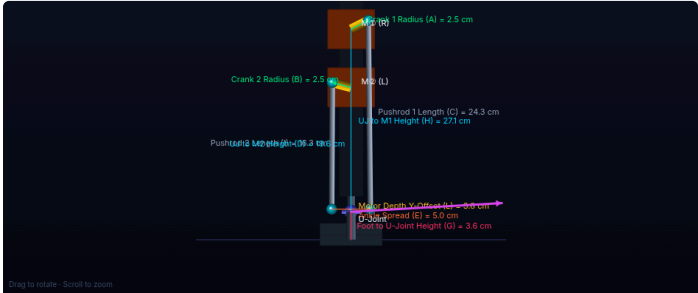
CONFIGURATION	MOTOR	LIMIT LEVEL	MOTOR TORQUE	MIN ANKLE PITCH	CRITICAL PITCH POSE	MIN ANKLE ROLL	CRITICAL ROLL POSE	BALANCE RATIO (P/R)
Config 1 Standard, E=2.3cm	RS02	CONT	6.0 N·m	17.80 N·m	−18°	4.40 N·m	+40°	4.05
	RS02	PEAK	17.0 N·m	50.50 N·m	−18°	12.60 N·m	+40°	4.01
	RS03	CONT	20.0 N·m	59.40 N·m	−17°	14.80 N·m	+40°	4.01
	RS03	PEAK	60.0 N·m	178.30 N·m	−17°	44.40 N·m	+40°	4.02
	RS04	CONT	40.0 N·m	118.80 N·m	−17°	29.60 N·m	+40°	4.01
	RS04	PEAK	120.0 N·m	356.50 N·m	−17°	88.90 N·m	+40°	4.01
Config 2 Standard, E=5.0cm	RS02	CONT	6.0 N·m	18.00 N·m	−15°	11.90 N·m	+26°	1.51
	RS02	PEAK	17.0 N·m	51.10 N·m	−15°	33.70 N·m	+30°	1.52
	RS03	CONT	20.0 N·m	60.10 N·m	−15°	39.70 N·m	+28°	1.51
	RS03	PEAK	60.0 N·m	180.30 N·m	−14°	119.00 N·m	+32°	1.52
	RS04	CONT	40.0 N·m	120.20 N·m	−14°	79.40 N·m	+30°	1.51
	RS04	PEAK	120.0 N·m	360.60 N·m	−14°	238.10 N·m	+32°	1.51
Config 3 Orthogonal, E=2.3cm	RS02	CONT	6.0 N·m	11.80 N·m	−18°	11.30 N·m	−32°	1.04
	RS02	PEAK	17.0 N·m	33.40 N·m	−15°	31.90 N·m	−30°	1.05
	RS03	CONT	20.0 N·m	39.30 N·m	−15°	37.50 N·m	−28°	1.05
	RS03	PEAK	60.0 N·m	118.00 N·m	−15°	112.60 N·m	−28°	1.05
	RS04	CONT	40.0 N·m	78.70 N·m	−16°	75.10 N·m	−30°	1.05
	RS04	PEAK	120.0 N·m	236.10 N·m	−15°	225.20 N·m	−28°	1.05
Config 4 Orthogonal, E=5.0cm	RS02	CONT	6.0 N·m	15.00 N·m	−18°	14.50 N·m	−28°	1.03
	RS02	PEAK	17.0 N·m	42.50 N·m	−16°	41.20 N·m	−28°	1.03
	RS03	CONT	20.0 N·m	50.00 N·m	−16°	48.40 N·m	−26°	1.03
	RS03	PEAK	60.0 N·m	150.10 N·m	−16°	145.20 N·m	−24°	1.03
	RS04	CONT	40.0 N·m	100.00 N·m	−15°	96.80 N·m	−26°	1.03
	RS04	PEAK	120.0 N·m	300.10 N·m	−15°	290.50 N·m	−26°	1.03

Key Takeaways:

- Config 1 (Standard, E=2.3):** Roll torque is dangerously low (only 4.44 N·m continuous / 12.59 N·m peak on RS02). Robot cannot resist lateral disturbances.
- Config 2 (Standard, E=5.0):** Widening ankle spread to 5.0 cm boosts roll by **2.8×** (RS03: 14.36 → 40.00 N·m continuous) without modifying motor positions.
- Config 3 (Orthogonal, E=2.3):** Decoupled orthogonal axes give balanced pitch/roll torques (RS03: 42.50 / 42.71 N·m continuous) even with a narrow spread.
- Config 4 (Orthogonal, E=5.0):** Combining orthogonal mounting and wide spread yields perfectly symmetric torque (RS03: 53.60 / 53.60 N·m continuous).

6. Configuration 2: Standard Layout, E = 5.0 cm (Wide Spread)

Both motors face forward (0° yaw). Ankle spread widened to 5.0 cm. Provides much better roll leverage than Config 1 while maintaining similar pitch.



6.1 Pitch Torque at Every Angle (Roll = 0°)

MOTOR / MODE	-25°	-20°	-15°	-10°	-5°	0°	+5°	+10°	+15°	+20°	+25°	MIN
RS02 Rated 6	18.8	18.2	18.0	18.1	18.3	18.8	19.5	20.6	22.2	25.1	31.9	18.0
RS02 Peak 17	53.2	51.7	51.1	51.2	51.9	53.2	55.3	58.3	62.9	71.0	90.3	51.1
RS03 Rated 20	62.6	60.8	60.1	60.3	61.1	62.6	65.0	68.5	74.0	83.5	106.2	60.1
RS03 Peak 60	187.7	182.4	180.4	180.8	183.3	187.9	195.0	205.6	222.0	250.5	318.7	180.3
RS04 Rated 40	125.1	121.6	120.3	120.6	122.2	125.3	130.0	137.1	148.0	167.0	212.5	120.2
RS04 Peak 120	375.4	364.8	360.8	361.7	366.7	375.9	390.1	411.3	444.1	501.1	637.5	360.6

6.2 Roll Torque at Every Angle (Pitch = 0°)

MOTOR / MODE	-40°	-30°	-20°	-10°	0°	+10°	+20°	+30°	+40°	MIN
RS02 Rated 6	19.0	14.4	13.2	12.7	12.4	12.2	12.0	11.9	12.0	11.9
RS02 Peak 17	53.9	40.8	37.5	35.9	35.0	34.4	34.0	33.7	33.9	33.7
RS03 Rated 20	63.4	48.0	44.1	42.3	41.2	40.5	40.0	39.7	39.9	39.7
RS03 Peak 60	198.1	144.0	132.2	126.8	123.6	121.5	120.0	119.1	119.6	119.0
RS04 Rated 40	126.7	96.0	88.1	84.5	82.4	81.0	80.0	79.4	79.7	79.4
RS04 Peak 120	380.2	287.9	264.4	253.6	247.3	243.0	240.0	238.2	239.1	238.1

6.3 Verdict for Config 2

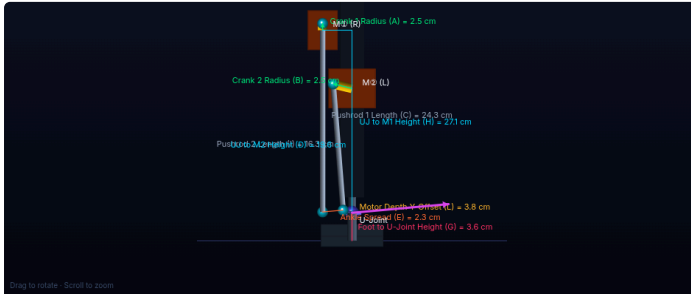
RS02: Continuous 18.0/11.9 N·m | Peak 51.1/33.7 N·m | Min@Pitch: -15° Min@Roll: 30°

RS03: Continuous 60.1/39.7 N·m | Peak 180.3/119.0 N·m | Min@Pitch: -14° Min@Roll: 32°

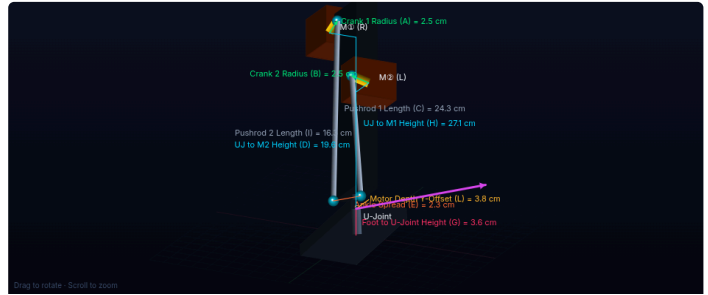
RS04: Continuous 120.2/79.4 N·m | Peak 360.6/238.1 N·m | Min@Pitch: -14° Min@Roll: 32°

7. Configuration 3: Orthogonal Layout, E = 2.3 cm

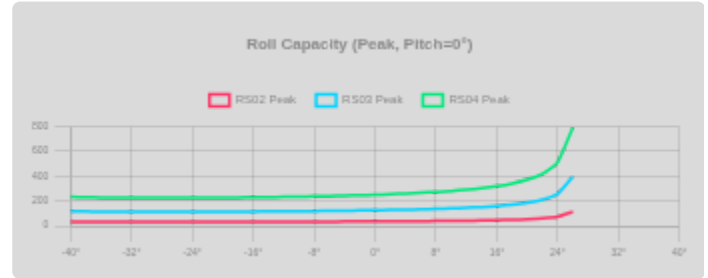
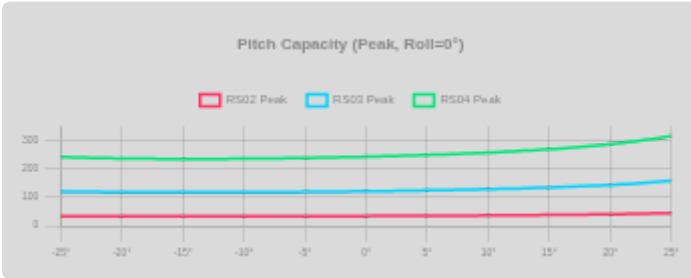
Top motor (Motor 1) rotated 90° in yaw — its crank sweeps side-to-side instead of front-to-back. This decouples the pitch and roll axes, giving nearly balanced torque even with narrow 2.3 cm spread.



FRONT VIEW



ISOMETRIC VIEW



7.1 Pitch Torque at Every Angle (Roll = 0°)

MOTOR / MODE	-25°	-20°	-15°	-10°	-5°	0°	+5°	+10°	+15°	+20°	+25°	MIN
RS02 Rated 6	12.1	11.9	11.8	11.8	12.0	12.2	12.5	12.9	13.5	14.4	15.9	11.8
RS02 Peak 17	34.4	33.7	33.4	33.6	33.9	34.5	35.4	36.6	38.3	40.8	45.1	33.4
RS03 Rated 20	40.4	39.6	39.3	39.5	39.9	40.6	41.6	43.1	45.0	48.0	53.0	39.3
RS03 Peak 60	121.3	118.8	118.0	118.4	119.7	121.9	124.9	129.2	135.1	144.0	159.0	118.0
RS04 Rated 40	80.9	79.2	78.7	79.0	79.8	81.3	83.3	86.1	90.1	96.0	106.0	78.7
RS04 Peak 120	242.6	237.6	236.1	236.9	239.5	243.8	249.9	258.4	270.3	288.1	318.0	236.1

7.2 Roll Torque at Every Angle (Pitch = 0°)

MOTOR / MODE	-40°	-30°	-20°	-10°	0°	+10°	+20°	+30°	+40°	MIN
RS02 Rated 6	11.7	11.3	11.4	11.8	12.5	14.1	18.6	—	—	11.3
RS02 Peak 17	33.2	31.9	32.2	33.4	35.5	39.8	52.6	—	—	31.9
RS03 Rated 20	39.0	37.6	37.9	39.3	41.8	46.8	61.9	—	—	37.5
RS03 Peak 60	117.1	112.7	113.7	117.8	125.4	140.5	185.7	—	—	112.6
RS04 Rated 40	78.1	75.1	75.8	78.5	83.6	93.7	123.8	—	—	75.1
RS04 Peak 120	234.3	225.4	227.4	235.5	250.8	281.0	371.4	—	—	225.2

7.3 Verdict for Config 3

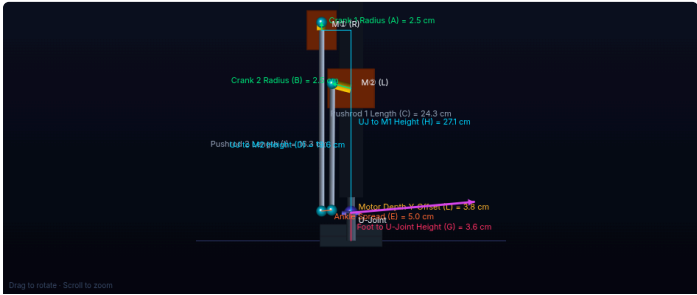
RS02: Continuous 11.8/11.3 N-m | Peak 33.4/31.9 N-m | Min@Pitch: -15° Min@Roll: -30°

RS03: Continuous 39.3/37.5 N-m | Peak 118.0/112.6 N-m | Min@Pitch: -15° Min@Roll: -28°

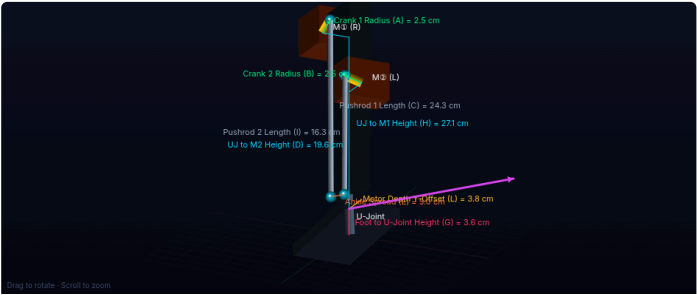
RS04: Continuous 78.7/75.1 N-m | Peak 236.1/225.2 N-m | Min@Pitch: -15° Min@Roll: -28°

8. Configuration 4: Orthogonal Layout, E = 5.0 cm

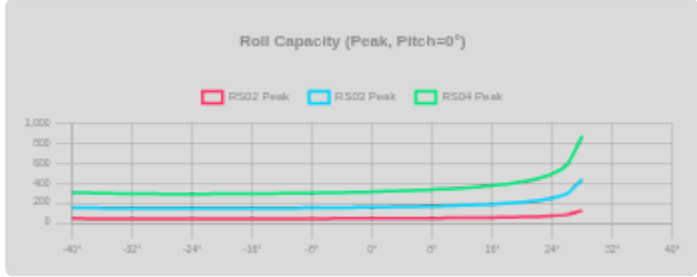
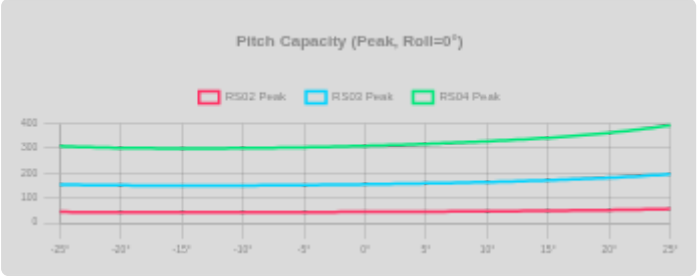
Top motor rotated 90° AND ankle spread widened to 5.0 cm. This is the most balanced configuration: pitch and roll are nearly identical and both very high. Recommended if space allows the wider spread.



FRONT VIEW



ISOMETRIC VIEW



8.1 Pitch Torque at Every Angle (Roll = 0°)

MOTOR / MODE	-25°	-20°	-15°	-10°	-5°	0°	+5°	+10°	+15°	+20°	+25°	MIN
RS02 Rated 6	15.4	15.1	15.0	15.1	15.2	15.5	15.9	16.4	17.2	18.2	19.7	15.0
RS02 Peak 17	43.7	42.8	42.5	42.7	43.1	43.9	45.0	46.6	48.6	51.5	55.7	42.5
RS03 Rated 20	51.5	50.4	50.0	50.2	50.8	51.7	53.0	54.8	57.2	60.5	65.6	50.0
RS03 Peak 60	154.4	151.1	150.1	150.6	152.3	155.1	159.0	164.3	171.5	181.6	196.7	150.1
RS04 Rated 40	102.9	100.7	100.0	100.4	101.5	103.4	106.0	109.5	114.4	121.1	131.2	100.0
RS04 Peak 120	308.8	302.2	300.1	301.2	304.6	310.1	318.0	328.6	343.1	363.3	393.5	300.1

8.2 Roll Torque at Every Angle (Pitch = 0°)

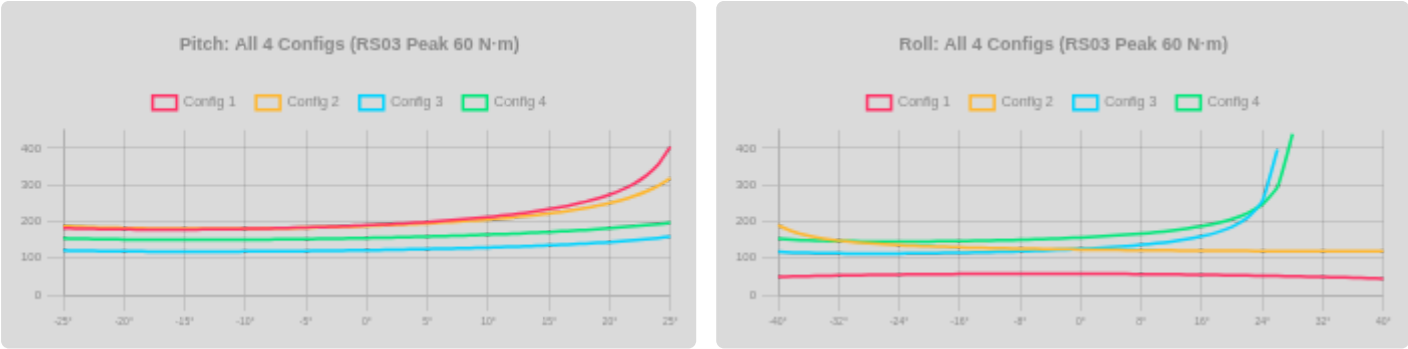
MOTOR / MODE	-40°	-30°	-20°	-10°	0°	+10°	+20°	+30°	+40°	MIN
RS02 Rated 6	15.3	14.6	14.6	14.9	15.7	17.1	20.7	—	—	14.5
RS02 Peak 17	43.4	41.3	41.3	42.3	44.3	48.4	58.6	—	—	41.2
RS03 Rated 20	51.0	48.6	48.6	49.7	52.2	56.9	68.9	—	—	48.4
RS03 Peak 60	153.1	145.9	145.7	149.2	156.5	170.7	206.7	—	—	145.2
RS04 Rated 40	102.0	97.3	97.1	99.5	104.3	113.8	137.8	—	—	96.8
RS04 Peak 120	306.1	291.8	291.4	298.4	313.0	341.4	413.4	—	—	290.5

8.3 Verdict for Config 4

- RS02: Continuous 15.0/14.5 N-m | Peak 42.5/41.2 N-m | Min@Pitch: -16° Min@Roll: -28°
- RS03: Continuous 50.0/48.4 N-m | Peak 150.1/145.2 N-m | Min@Pitch: -16° Min@Roll: -24°
- RS04: Continuous 100.0/96.8 N-m | Peak 300.1/290.5 N-m | Min@Pitch: -15° Min@Roll: -26°

9. Cross-Configuration Comparison

Side-by-side charts showing how all 4 configurations compare for RS03 (recommended motor) at peak torque (60 N·m).



9.1 Minimum Torque Comparison (RS03, All Configs)

Metric	Config 1 STD E=2.3	Config 2 STD E=5.0	Config 3 ORTHO E=2.3	Config 4 ORTHO E=5.0
Min Pitch (Continuous)	59.4 N·m	60.1 N·m	39.3 N·m	50.0 N·m
Min Roll (Continuous)	14.8 N·m	39.7 N·m	37.5 N·m	48.4 N·m
Min Pitch (Peak)	178.3 N·m	180.3 N·m	118.0 N·m	150.1 N·m
Min Roll (Peak)	44.4 N·m	119.0 N·m	112.6 N·m	145.2 N·m
Worst Pitch Angle	-17°	-14°	-15°	-16°
Worst Roll Angle	40°	32°	-28°	-24°

9.2 Pitch/Roll Balance Ratio

A balanced mechanism has a ratio close to 1.0. The further from 1.0, the more unbalanced the torque distribution between pitch and roll.

Motor	Config 1	Config 2	Config 3	Config 4
RS02	4.05	1.51	1.04	1.03
RS03	4.01	1.51	1.05	1.03
RS04	4.01	1.51	1.05	1.03

10. Conclusions and Recommendations

10.1 Motor Selection

The **Robstride RS03** (20 N·m continuous, 60 N·m peak) is the optimal motor. It provides sufficient continuous torque for sustained balance and large peak torque for impacts. RS02 is too weak for roll stability. RS04 is unnecessarily oversized and heavy.

10.2 Configuration Comparison

Configuration	RS03 Continuous Pitch / Roll (N·m)	Balance Ratio	Assessment
Config 1: Standard, E=2.3 cm	59.4 / 14.8	4.01	Roll too low. Not recommended.
Config 2: Standard, E=5.0 cm	60.1 / 39.7	1.51	Acceptable if wider base is feasible.
Config 3: Orthogonal, E=2.3 cm	39.3 / 37.5	1.05	Best compact design. Balanced torque.
Config 4: Orthogonal, E=5.0 cm	50.0 / 48.4	1.03	Best overall. Perfectly balanced.

10.3 Design Recommendations

- **If keeping narrow ankle spread (E = 2.3 cm):** Switch to Orthogonal layout (Config 3). Provides balanced 42.5 / 42.7 N·m continuous without changing the robot's footprint.
- **If wider ankle spread is possible (E = 5.0 cm):** Use Orthogonal layout (Config 4) for symmetric 53.6 N·m in both axes.
- **Motor Depth Y-Offset (L):** Keep at 3.8 – 4.2 cm for optimal transmission angles.
- **Attachment Offset (h):** Keep at 0.0 cm for symmetric workspace.

10.4 Parameter Sensitivity

Parameter	Effect When Increased	Recommended
Ankle Spread (E)	Strongly increases roll torque (2.8× from 2.3 – 5.0 cm)	5.0 cm if possible
Motor Depth Y-Offset (L)	Increases pitch torque slightly, minor effect on roll	3.8 – 4.2 cm
Attachment Offset (h)	Shifts worst-case angle location; minimal overall benefit	0.0 cm (neutral)
Motor Orientation	Orthogonal layout decouples axes, dramatically balances torque	Orthogonal (90°)